WORKED SOLUTION

STEP I 2012, Question 7

Pure • Periodic Recurrence Relations

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/12-s1-q7

The key idea

A sequence that repeats with period 1, 2 or 3 is pinned down by its first few terms, so the recurrence only has to hold for the few consecutive triples inside one period (plus the wrap-around). Writing these out gives a small set of simultaneous linear equations in x, y, z. The equations have a solution that isn't all zeros, and that is what forces the condition on p and q. Throughout, the fact that $x \neq 0$ is what lets us cancel.

Solution

Part (i)

If $t_n = x$ for every n, the recurrence says $x = px + qx$, that is $(p+q-1)x = 0$. Since $x \neq 0$, we get $p + q = 1$.

Conversely, if $p + q = 1$ and $t_0 = t_1 = x$ then $t_2 = (p+q)x = x$, and by induction every term equals x. Nothing restricts x, so it can be any non-zero real number.

Part (ii)

The pattern $x, y, x, y, \ldots$ satisfies the recurrence exactly when the two consecutive triples (x, y, x) and (y, x, y) do:

$$x = py + qx, \qquad \text{i.e.} \quad (1-q)x = py, \tag{1}$$

$$y = px + qy, \qquad \text{i.e.} \quad (1-q)y = px. \tag{2}$$

Adding and subtracting (1) and (2) gives

$$(1-q-p)(x+y) = 0, \qquad (1-q+p)(x-y) = 0.$$

The numbers $x + y$ and $x - y$ can't both be zero, since that would make $x = 0$. If $x + y \neq 0$ then $p + q = 1$; if $x - y \neq 0$ then $q - p = 1$. So at least one of $q + p = 1$ and $q - p = 1$ holds, as required.

Now we deduce the relation between x and y.

- If $q + p = 1$ then $1 - q = p$ and (1) becomes $px = py$. So $x = y$, *provided* $p \neq 0$.
- If $q - p = 1$ then $1 - q = -p$ and (1) becomes $-px = py$. So $x = -y$, again provided $p \neq 0$.

The only escape is $p = 0$, and then either case gives $q = 1$. With $p = 0$ and $q = 1$ the recurrence is $t_{n+2} = t_n$, which every pattern $x, y, x, y, \ldots$ satisfies, so x and y are unrelated. Hence $x = y$ or $x = -y$ unless $p = 0$ and $q = 1$.

Part (iii)

Now the pattern is $x, y, z, x, y, z, \ldots$, and we need the three consecutive triples (x, y, z), (y, z, x) and (z, x, y):

$$z = py + qx, \tag{3}$$
$$x = pz + qy, \tag{4}$$
$$y = px + qz. \tag{5}$$

We use (3) to eliminate z from the other two:

$$\text{(4):}\quad x = p(py + qx) + qy \quad\Longrightarrow\quad (pq - 1)x + (p^2 + q)y = 0,$$
$$\text{(5):}\quad y = px + q(py + qx) \quad\Longrightarrow\quad (p + q^2)x + (pq - 1)y = 0.$$

Multiply the first equation by $pq - 1$ and the second by $p^2 + q$, then subtract. The y terms cancel, leaving

$$\left[(pq - 1)^2 - (p^2 + q)(p + q^2)\right]x = 0.$$

Ask yourself Why must the bracket vanish? Couldn't the equations simply have no solution?

They do have a solution: the sequence exists, so x and y satisfy both equations. And $x \neq 0$, so we can divide by it. Expanding the bracket,

$$\begin{aligned}(pq - 1)^2 - (p^2 + q)(p + q^2) &= p^2q^2 - 2pq + 1 - \left(p^3 + p^2q^2 + pq + q^3\right)\\ &= 1 - 3pq - p^3 - q^3,\end{aligned}$$

so $p^3 + q^3 + 3pq - 1 = 0$, as required.

Factorising. The expression is $a^3 + b^3 + c^3 - 3abc$ with $a = p$, $b = q$, $c = -1$, and the standard identity

$$a^3 + b^3 + c^3 - 3abc = (a + b + c) \cdot \tfrac{1}{2}\left[(a - b)^2 + (b - c)^2 + (c - a)^2\right]$$

gives

$$p^3 + q^3 + 3pq - 1 = \tfrac{1}{2}(p + q - 1)\left[(p - q)^2 + (q + 1)^2 + (p + 1)^2\right].$$

(If you don't remember the identity, check it by expanding: the square bracket is $2(p^2+q^2+1-pq+p+q)$, and multiplying by $p+q-1$ gives back the cubic.) The product is zero, so either $p+q=1$ or $(p-q)^2+(p+1)^2+(q+1)^2=0$.

Case $(p-q)^2+(p+1)^2+(q+1)^2=0$**.** A sum of real squares is zero only if each square is zero, so $p=q=-1$. Then (3) reads $z=-y-x$, that is, $x+y+z=0$.

Case $p+q=1$**.** Substitute $q=1-p$ into (3), (4) and (5) and rearrange each as a difference:

$$z-x=p(y-x), \qquad x-y=p(z-y), \qquad y-z=p(x-z).$$

Write $a=y-x$, $b=z-y$, $c=x-z$, so that $a+b+c=0$. The first and third equations say $c=-pa$ and $b=-pc=p^2a$. Then

$$0=a+b+c=a(1-p+p^2).$$

? **Ask yourself** Can $1-p+p^2$ be zero?

No: $1-p+p^2=\left(p-\frac{1}{2}\right)^2+\frac{3}{4}>0$ for every real p. So $a=0$, which gives $c=-pa=0$ and then $b=0$. That is $x=y=z$.

The two cases don't overlap, because $p=q=-1$ gives $p+q=-2\neq 1$. So either $x=y=z$ (when $p+q=1$) or $x+y+z=0$ (when $p=q=-1$). Both happen: in the second case the recurrence $t_{n+2}=-t_{n+1}-t_n$ gives $x,\ y,\ -x-y,\ x,\ y,\ \ldots$ for any starting values.

Checks

- With $p=q=-1$, $t_0=2$ and $t_1=5$, the sequence runs $2,5,-7,2,5,-7,\ldots$, and $2+5-7=0$.
- With $p=3$ and $q=4$ (so $q-p=1$), starting $1,-1$ gives $t_2=-3+4=1$ and $t_3=3-4=-1$, alternating as part (ii) predicts.
- At $p=2$, $q=-1$ the cubic gives $8-1-6-1=0$, consistent with $p+q=1$. We also checked the expansion, the factorisation and the cubic's agreement with the 3×3 determinant of (3)–(5) at 1000 random (p,q). Simulating the recurrence confirmed every case above.

Common slips

- Concluding $p+q=1$ *and* $q-p=1$ in part (ii). The argument shows only that at least one holds; both hold only in the exceptional case $p=0$, $q=1$.
- Dividing by p in part (ii) without noticing that $p=0$ is the exception the question is asking for.
- Forgetting the wrap-around equations (4) and (5). One equation per period isn't enough to force the condition.

Beyond STEP

The standard way to handle $t_{n+2} = pt_{n+1} + qt_n$ is through the *characteristic equation* $\lambda^2 = p\lambda + q$: in general $t_n = A\alpha^n + B\beta^n$, where α and β are its roots. A sequence repeats every k steps exactly when the roots it uses satisfy $\lambda^k = 1$. Constant sequences need the root 1 (so $p + q = 1$), and alternating ones need -1 (so $q - p = 1$). The case $p = q = -1$ gives $\lambda^2 + \lambda + 1 = 0$, whose roots are the complex cube roots of unity, and that is why their sum $1 + \omega + \omega^2 = 0$ reappears as $x + y + z = 0$. This is linear algebra in disguise (the shift map acting on sequences), and Strang's *Introduction to Linear Algebra* treats recurrences through eigenvalues in exactly this way.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.